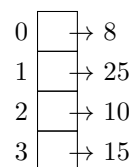


1 External Chaining

Consider the following External Chaining Hash Set below, which doubles in size when the load factor reaches 1.5. Assume that we're using the default hashCode for integers, which simply returns the integer itself.



- (a) Draw the External Chaining Hash Set that results if we insert 18.
- (b) Draw the External Chaining Hash Set that results if we insert 5 after the insertion done in part (a).

2 Invalid Hashes

For both parts below, suppose we are trying to hash the following class:

```
import java.util.Random;
class Point {
    private int x;
    private int y;
    private static count = 0;

    public Point(int x, int y) {
        this.x = x;
        this.y = y;
        count += 1;
    }
}
```

(a) Which of the hashCodes are invalid?

- (i) **public void** hashCode() {
 System.out.print(this.x + this.y);
}
- (ii) **public int** hashCode() {
 Random randomGenerator = new Random();
 return randomGenerator.nextInt(Int);
}
- (iii) **public int** hashCode() {
 return this.x + this.y;
}
- (iv) **public int** hashCode() {
 return 4;
}
- (v) **public int** hashCode() {
 return count;
}

(b) *Extra:* Suppose we know all the Points have x and y coordinates between 0 and 10, inclusive. Suggest a **good** hashCode method.

3 Hashing Gone Crazy

For this question, use the following TA class for reference.

```
public class TA {
    int charisma;
    String name;
    TA(String name, int charisma) {
        this.name = name;
        this.charisma = charisma;
    }
    @Override
    public boolean equals(Object o) {
        TA other = (TA) o;
        return other.name.charAt(0) == this.name.charAt(0);
    }
    @Override
    public int hashCode() {
        return charisma;
    }
}
```

Assume that the hashCode of a TA object returns charisma, and the equals method returns true if and only if two TA objects have the same first letter in their name.

Assume that the ECHashMap is a HashMap implemented with external chaining as depicted in lecture. The ECHashMap instance begins at size 4 and, for simplicity, does not resize. Draw the contents of map after the executing the insertions below:

```
ECHashMap<TA, Integer> map = new ECHashMap<>();
TA sohum = new TA("Sohum", 10);
TA vivant = new TA("Vivant", 20);
map.put(sohum, 1);
map.put(vivant, 2);

vivant.charisma += 2;
map.put(vivant, 3);

sohum.name = "Vohum";
map.put(vivant, 4);

sohum.charisma += 2;
map.put(sohum, 5);

sohum.name = "Sohum";
TA shubha = new TA("Shubha", 24);
map.put(shubha, 6);
```